

Rovnosti, které máme dokázat, si rozepíšeme tak, aby jejich správnost byla očividná.

$$\operatorname{Re}(\langle u, v \rangle) = \frac{1}{4} (\|u + v\|^2 - \|u - v\|^2)$$

$$4 \operatorname{Re}(\langle u, v \rangle) = \langle u + v, u + v \rangle - \langle u - v, u - v \rangle$$

$$4 \operatorname{Re}(\langle u, v \rangle) = \langle u, u \rangle + \langle v, u \rangle + \langle u, v \rangle + \langle v, v \rangle - (\langle u, u \rangle - \langle v, u \rangle - \langle u, v \rangle + \langle v, v \rangle)$$

$$4 \operatorname{Re}(\langle u, v \rangle) = \langle v, u \rangle + \langle u, v \rangle - (-\langle v, u \rangle - \langle u, v \rangle)$$

$$4 \operatorname{Re}(\langle u, v \rangle) = 2 \overline{\langle v, u \rangle} + 2 \langle u, v \rangle$$

$$2 \operatorname{Re}(\langle u, v \rangle) = \overline{\langle u, v \rangle} + \langle u, v \rangle$$

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u - iv\|^2 - \|u + iv\|^2)$$

$$4 \operatorname{Im}(\langle u, v \rangle) = \langle u - iv, u - iv \rangle - \langle u + iv, u + iv \rangle$$

$$4 \operatorname{Im}(\langle u, v \rangle) = \langle u, u \rangle - \langle iv, u \rangle - \langle u, iv \rangle + \langle iv, iv \rangle - (\langle u, u \rangle + \langle iv, u \rangle + \langle u, iv \rangle + \langle iv, iv \rangle)$$

$$4 \operatorname{Im}(\langle u, v \rangle) = -\langle iv, u \rangle - \langle u, iv \rangle - (\langle iv, u \rangle + \langle u, iv \rangle)$$

$$-2 \operatorname{Im}(\langle u, v \rangle) = \overline{\langle iv, u \rangle} + \langle u, iv \rangle$$

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$$-2 \operatorname{Im}(\langle u, v \rangle) = i \overline{\langle u, v \rangle} + i \langle u, v \rangle$$

$$-2 \operatorname{Im}(\langle u, v \rangle) = i (\langle u, v \rangle - \overline{\langle u, v \rangle})$$

$$2i \operatorname{Im}(\langle u, v \rangle) = \langle u, v \rangle - \overline{\langle u, v \rangle}$$

Tímto jsou rovnosti dokázány.